Refractive Errors

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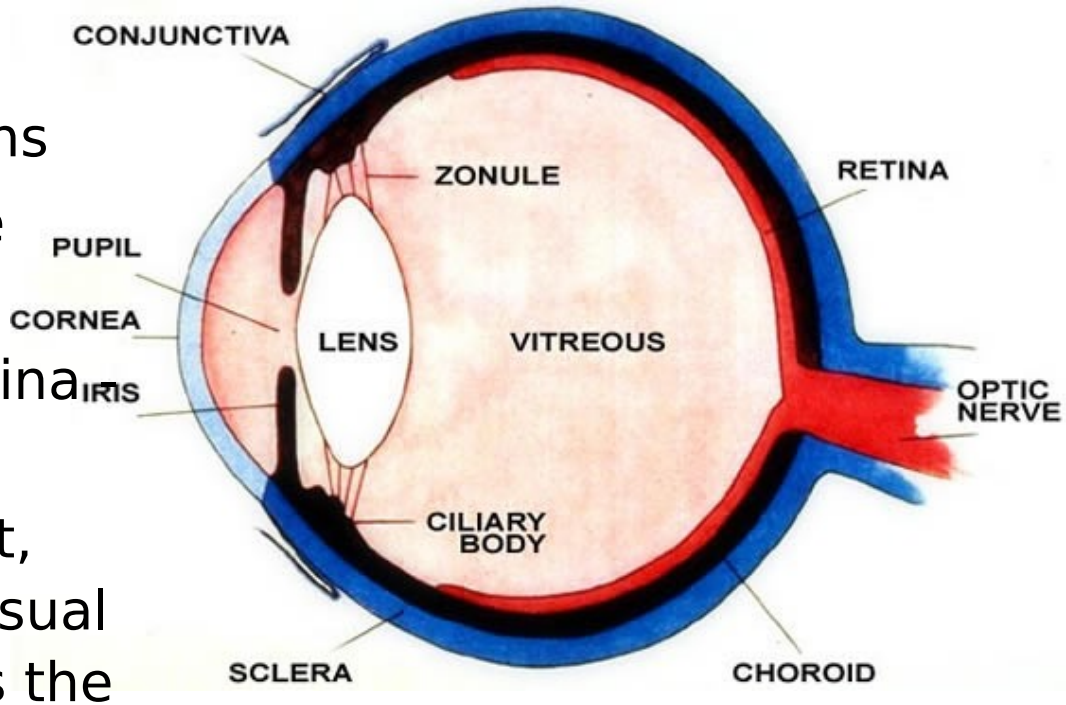
Year III - 2009

Objectives

- Understand normal vision
- Definitions
  - Myopia
  - Hyperopia
  - Astigmatism
  - Presbyopia
- Detection
- Treatment
  - Spectacle correction
  - Contact lenses
  - LASIK

Normal Vision

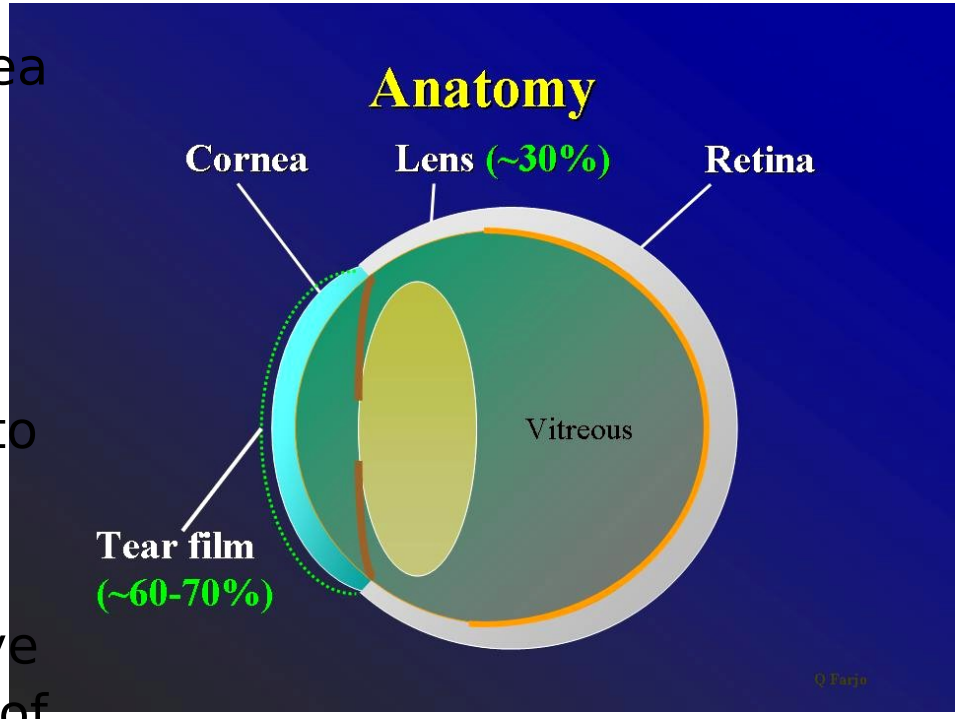
- Light eye thro’ eye cornea
- Then the pupil – regulation of light
- The light then hits the lens
- Next: passes through the vitreous humor
- Finally; it reaches the retina, image appears inverted
- ON carries signals of light, dark, and colors to the visual cortex - which assembles the signals into images (our vision)



Definitions

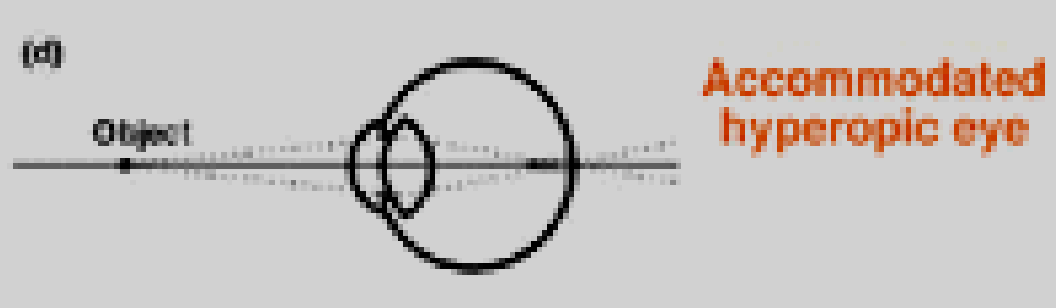
Refraction

- The ratio of the refractive power of the lens and cornea (refractive media) to the axial length of the globe
- Refractive power (RP) is specified in diopters
- RP is calculated according to laws of geometric optics
- Maximum total refractive power of an emmetropic eye is 63D with an axial length of 23.5 Cornea- 43D; lens – 20D



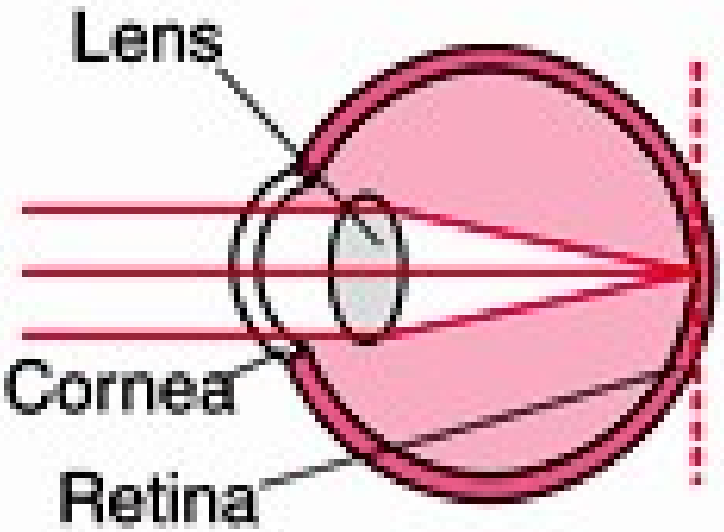
Definition.....

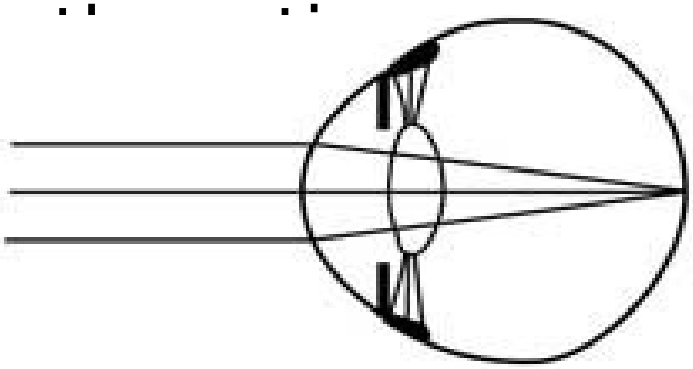
- Accommodation
  - The RP of the eye is not constant – alters for near and far
  - Involves
    - The lens,
    - Zonule fibres
    - Ciliary muscle



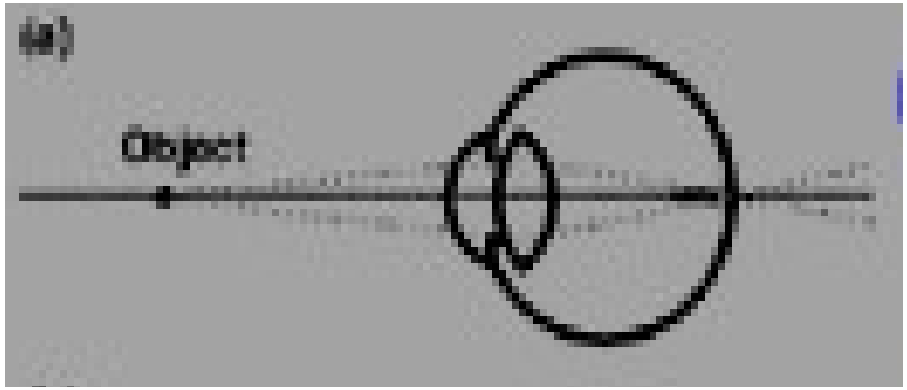
Emmetropia (normal sight)

- The ratio of refractive media:axial length is balanced
- Parallel rays meet at a focal point on





Normal eye

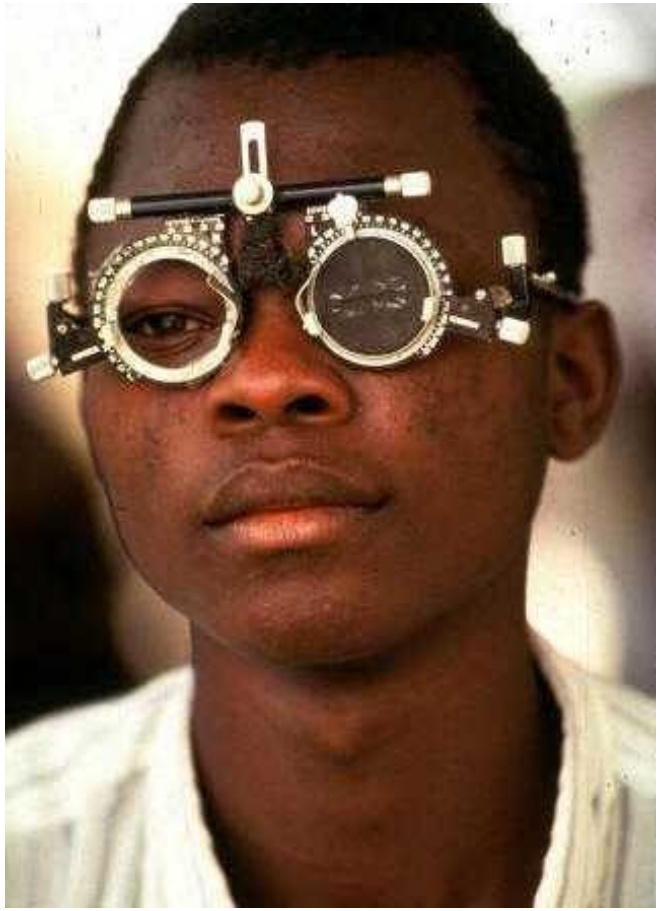


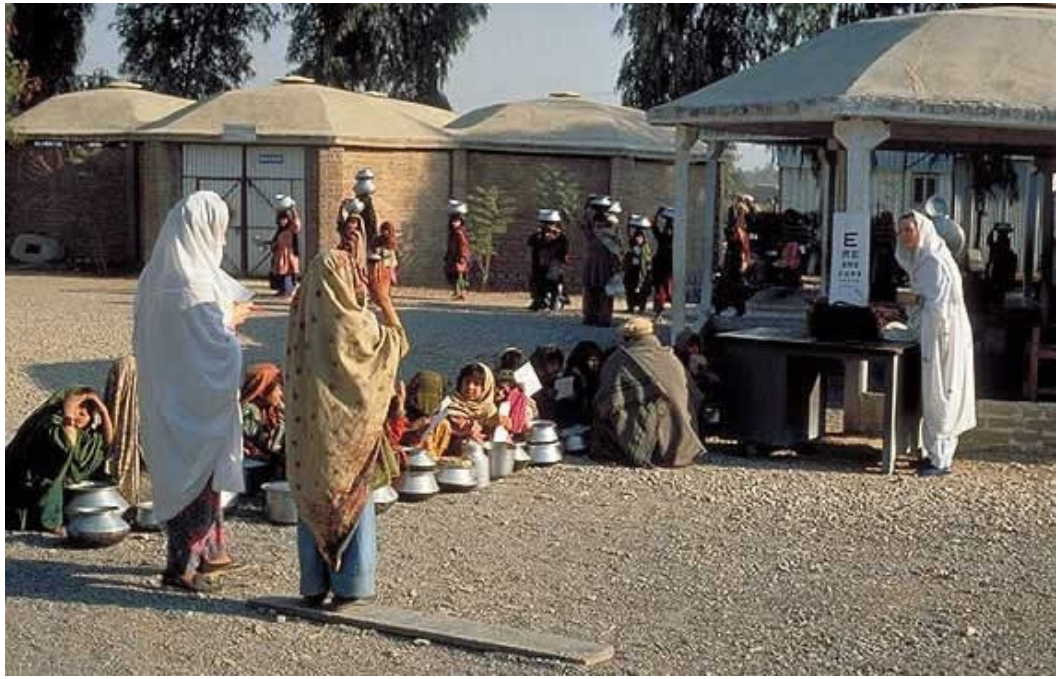
Ametropia (refractive error)

Definition:

- Mismatch between axial length of the eye and the refractive power of the cornea and the lens
  - Axial (common)
    - Myopia
    - Hypermetropia: (Presbyopia)
    - Astigmatism
  - Refractive
    - Cornea
    - Lenticular

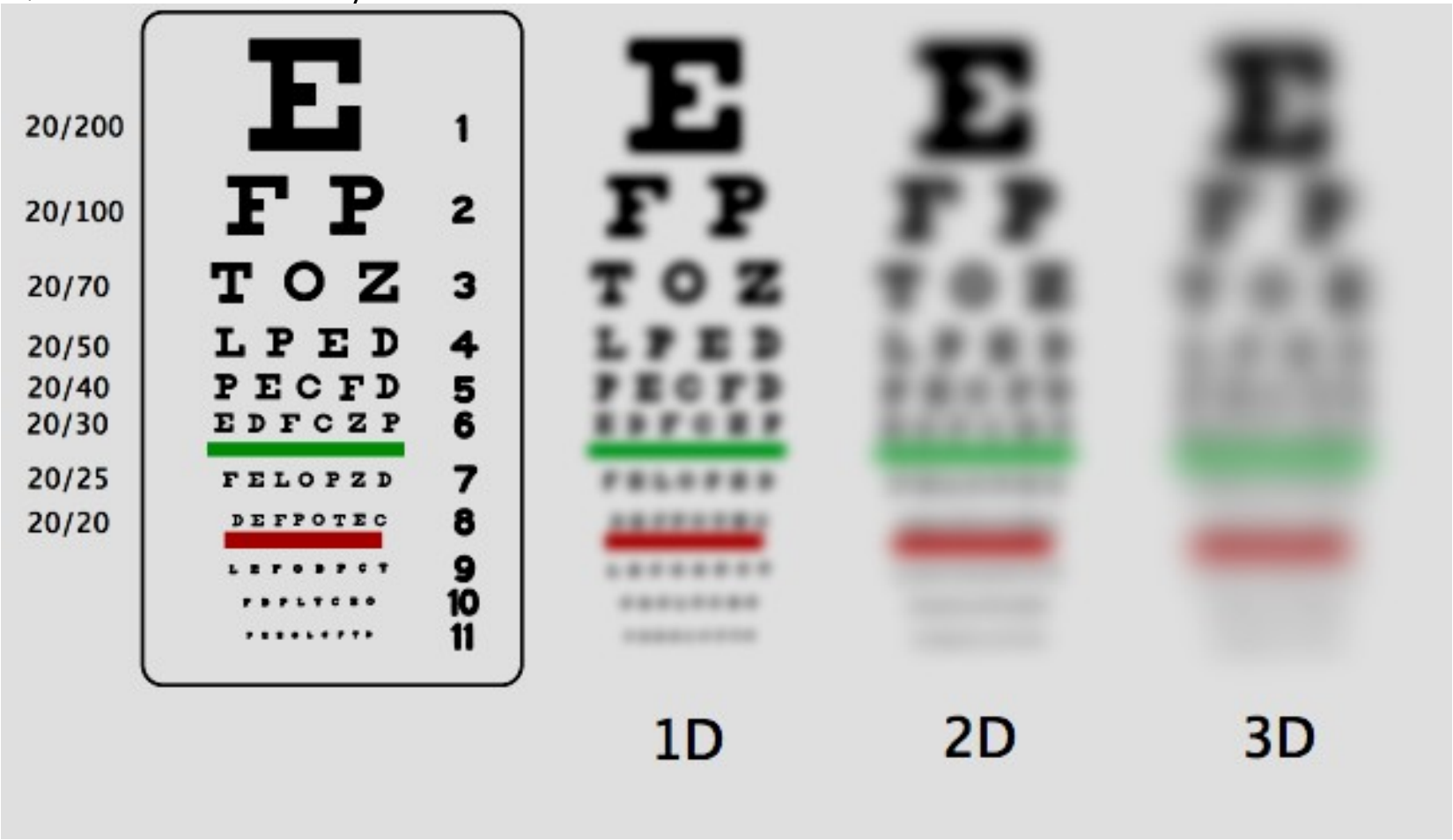
Testing Visual Acuity





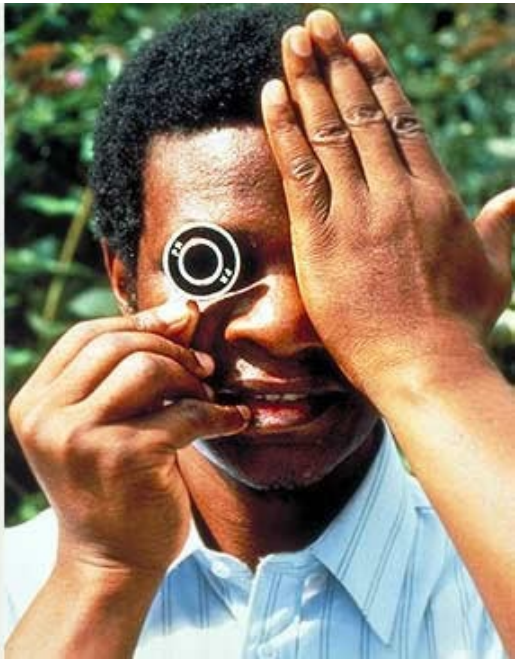
Examination

- Visual Acuity



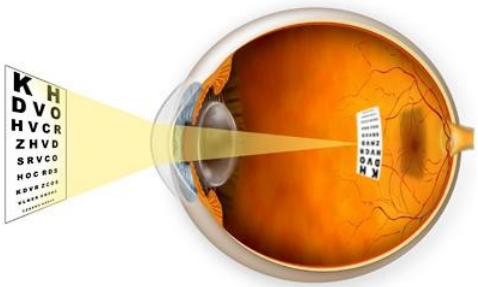
The Pin Hole Test

- Testing Distance Vision with the Pin-Hole
- Pinholes only allow the passage of light which is perpendicular to the lens, and thus does not need to be bent prior to being focused onto the retina



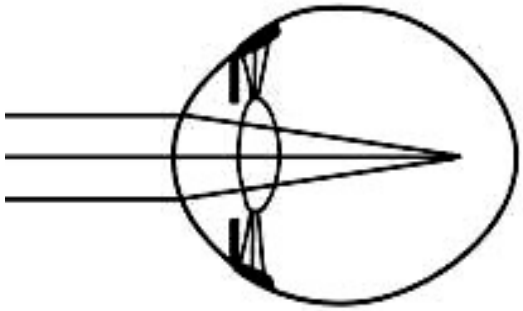
Snellen Notation Metric	Snellen Notation Imperial	MAR	Log MAR	Decimal
6/60	20/200	10	1.0	0.10
6/48	20/160	8.0	0.9	0.13
6/38	20/125	6.3	0.8	0.16
6/30	20/100	5.0	0.7	0.20
6/24	20/80	4.0	0.6	0.25
6/19	20/60	3.2	0.5	0.32
6/15	20/50	2.5	0.4	0.40
6/12	20/40	2.0	0.3	0.50
6/9.5	20/30	1.6	0.2	0.63
6/7.5	20/25	1.25	0.1	0.80
6/6	20/20	1.00	0.0	1.00
6/4.8	20/16	0.80	-0.1	1.25
6/3.8	20/12.5	0.63	-0.2	1.58
6/3.0	20/10	0.50	-0.3	2.00

Near Vision Testing

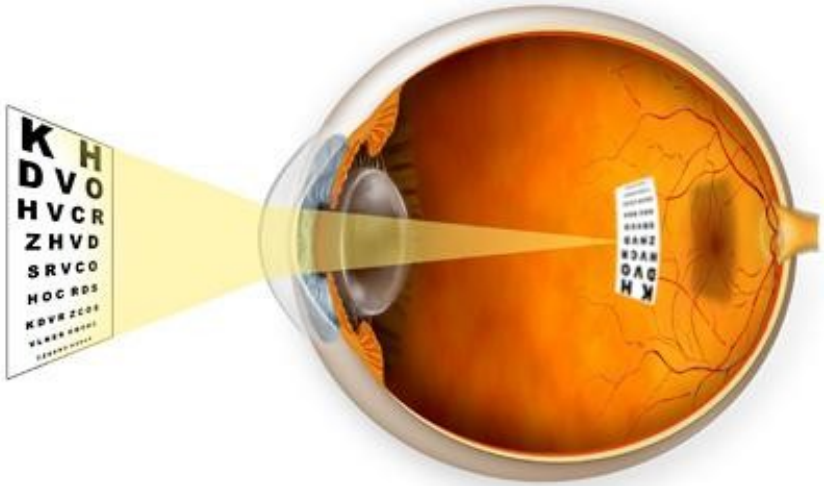


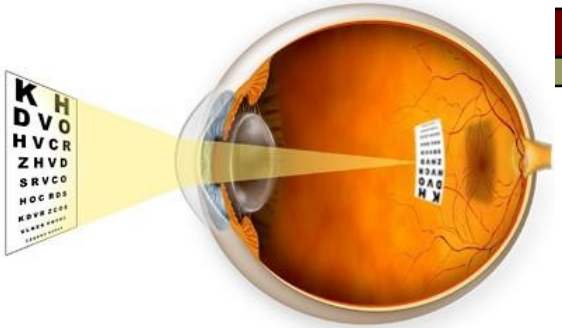
Myopia

- Definition
  - Discrepancy between the refractive power and the axial length – parallel rays converge at a focal point anterior to the retina



Light focused in front of retina

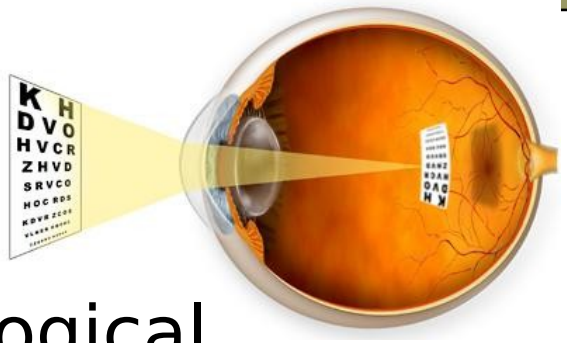




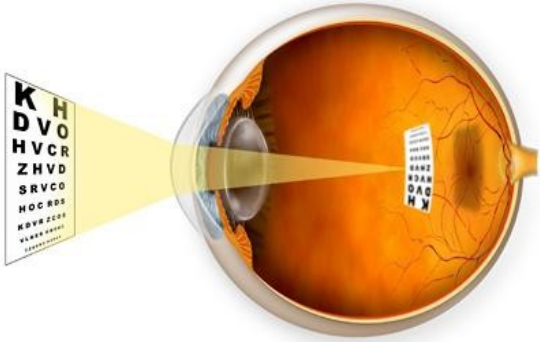
Myopia……

- Epidemiology
  - Generally: 25% of persons between 20 – 30 yrs (-1.0D)
  - Africa:
  - Nairobi (Kenya) - 10% 13-15yrs
  - Makueni (Kenya) – 2% 12-15 yrs

Myopia.....

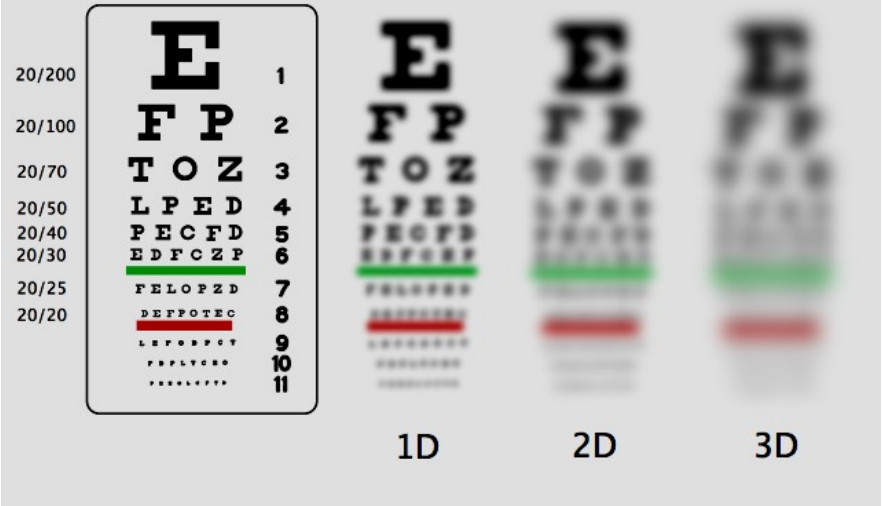


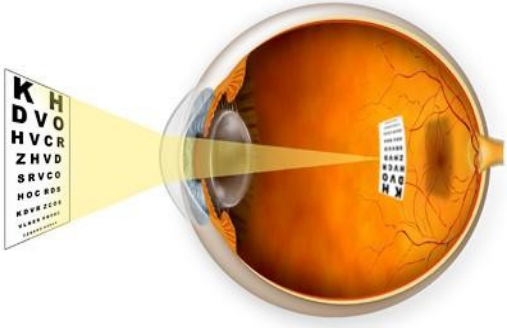
- Simple
  - Onset is 10 -12 years
  - Does not progress > 20 years (may have benign progressive – stabilizes at 30 yrs)
  - Refraction rarely > 6 Diopter
- Pathological
  - Largely hereditary
  - Varies in ethnic groups
  - Continuous progression
- Special forms of Myopia:
  - Lens-
    - Nuclear sclerosis
    - Spherophekia
  - Cornea
    - Keratoconus



Myopia……

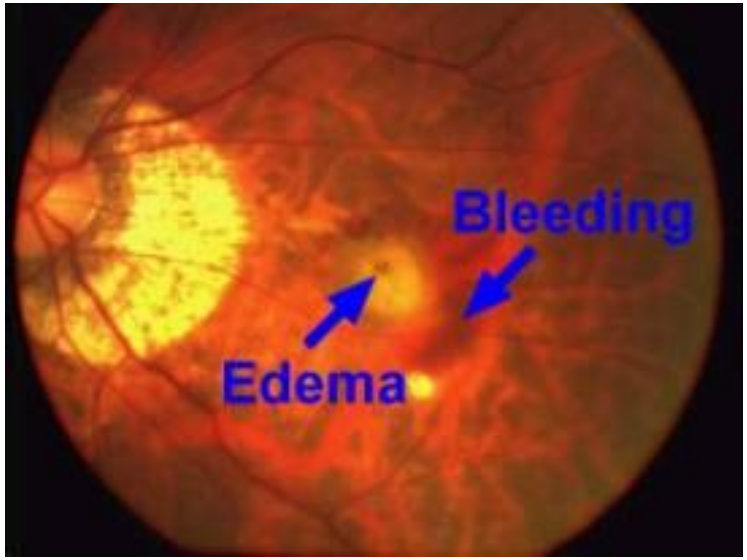
- Symptoms
  - Inability to see distant objects
  - “Squinting” to focus
  - Can read when without glasses (older myopes)
  - Simulation of esotropoia (shift in axes of the eye)
  - Floaters: Reduction in volume of vitreous, collapse, vitreous opacification, Increased risk of RD





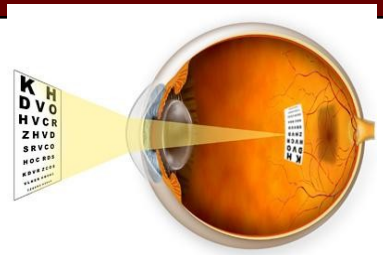
Myopia……

- Signs
  - Thinning of sclera
  - Deep anterior chamber
  - Atrophy of the ciliary muscle
  - Chorioretinal atrophy around the OD
  - Breaks in bruchs membrane
  - Subretinal neo-vascularization – retinal oedema/bleeding (Fuchs black sport)
  - Glaucoma
  - Optic nerve – oblique



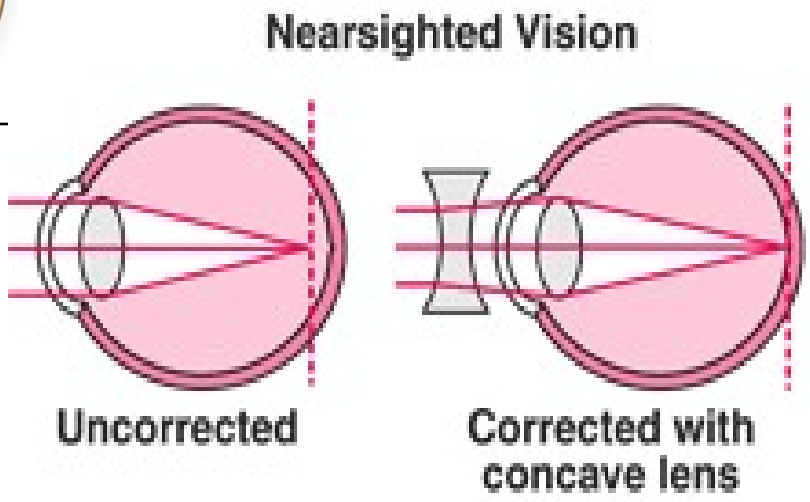
- Diagnosis
  - Refraction
    - Objective (Cycloplegic)
    - Subjective

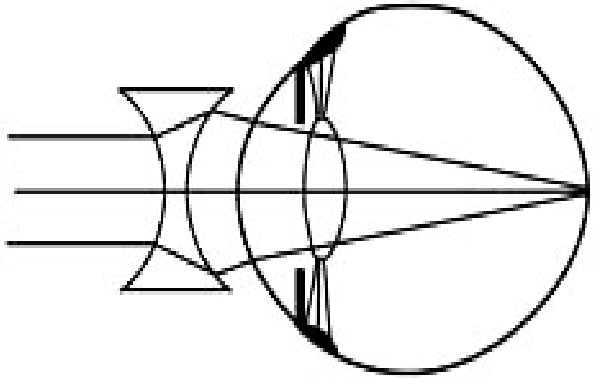
Myopia....

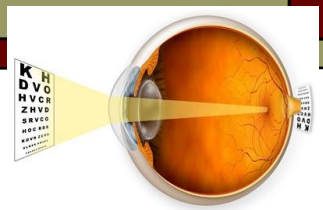


Treatment:

- Concave lenses
- Contact lenses
- Surgery
  - Removal of lens
  - LASIK

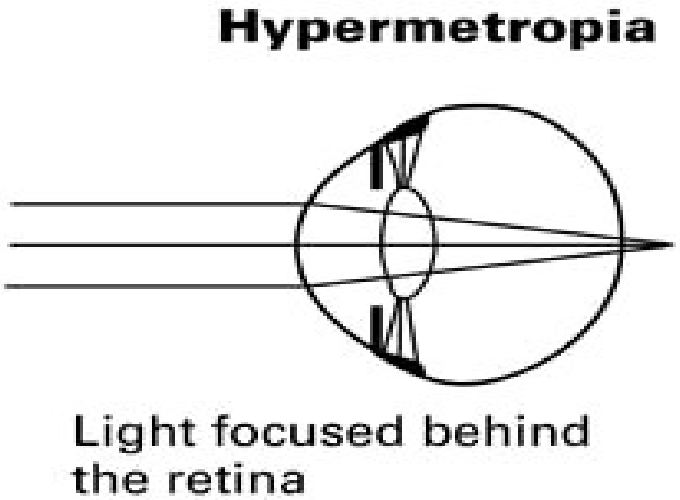


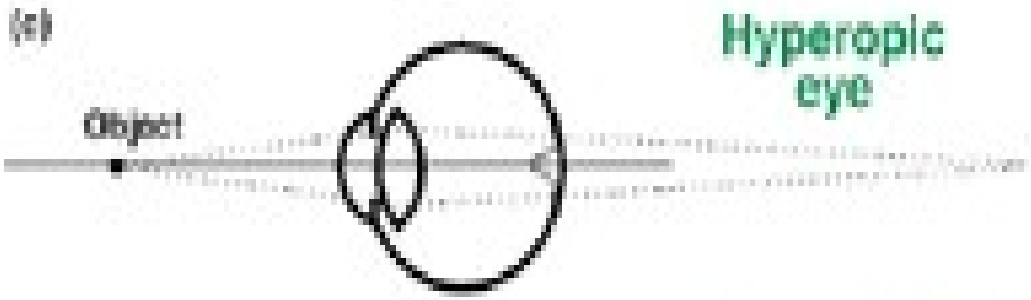


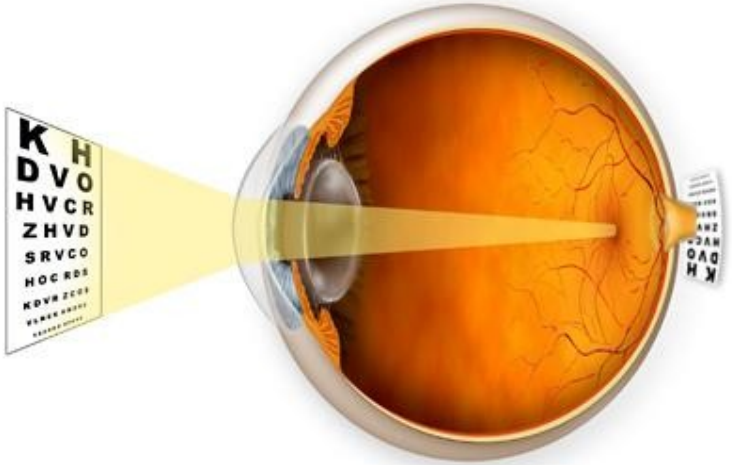


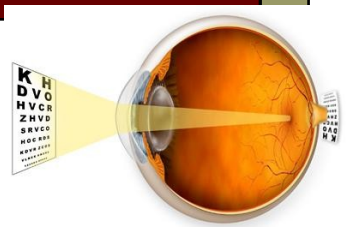
Hypermetropia…

- Definition
  - Discrepancy between the refractive power and the axial length – parallel rays converge at a focal point posterior to the



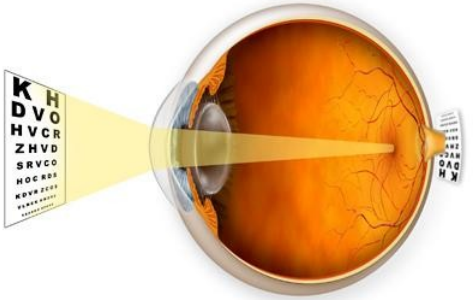






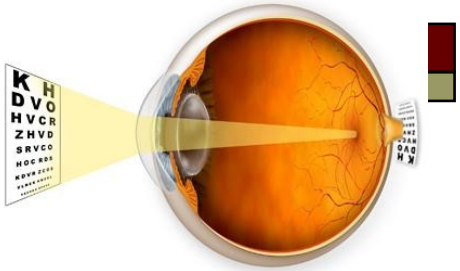
Hypermetropia.....

- Epidemiology
  - About 20% of ages 20-30yrs >+1.00DS
  - Most newborns
- Etiology
  - Exact mechanism not known
  - Special forms
    - Apakia: Congenital; Traumatic; Post Cataract Surgery



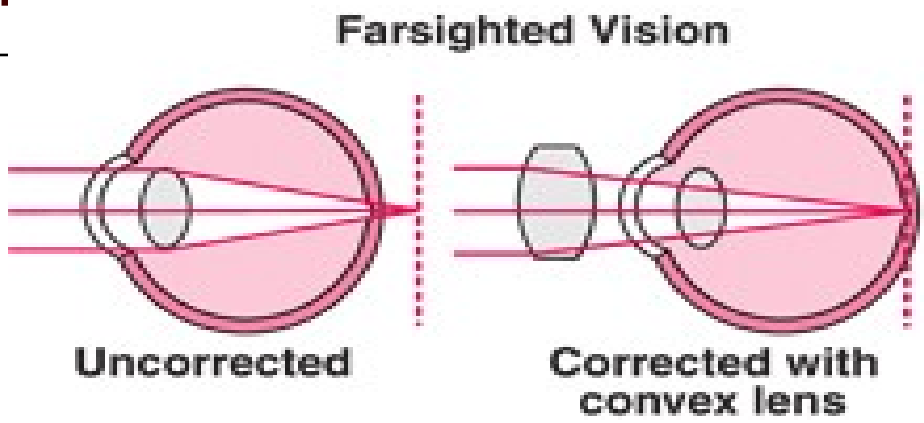
Hypermetropia.....

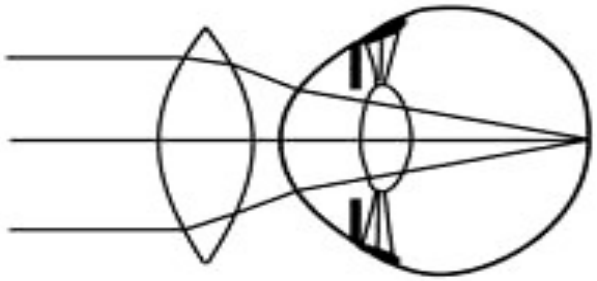
- Symptoms
  - asthenopic symptoms: eye pain; headache; Blurred vision; Fatigue
  - Esotropia
  - Early presbyopia
- Signs
  - Blurred disc
  - Tortuous retinal vessels
  - Latent hyperopia
  - refraction



Hypermetropia

- Treatment
  - Concave spectacles
  - Contact lenses
  - Surgery
    - LASIK

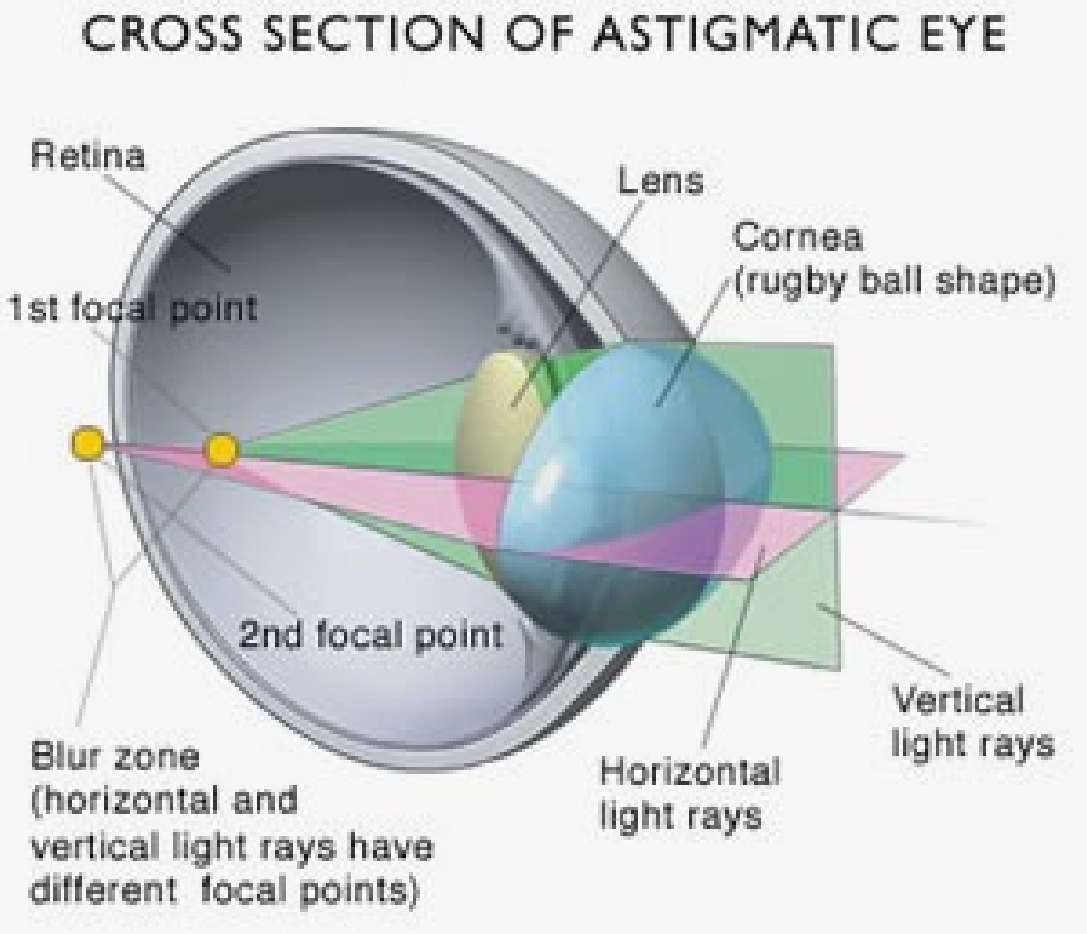




Astigmatism

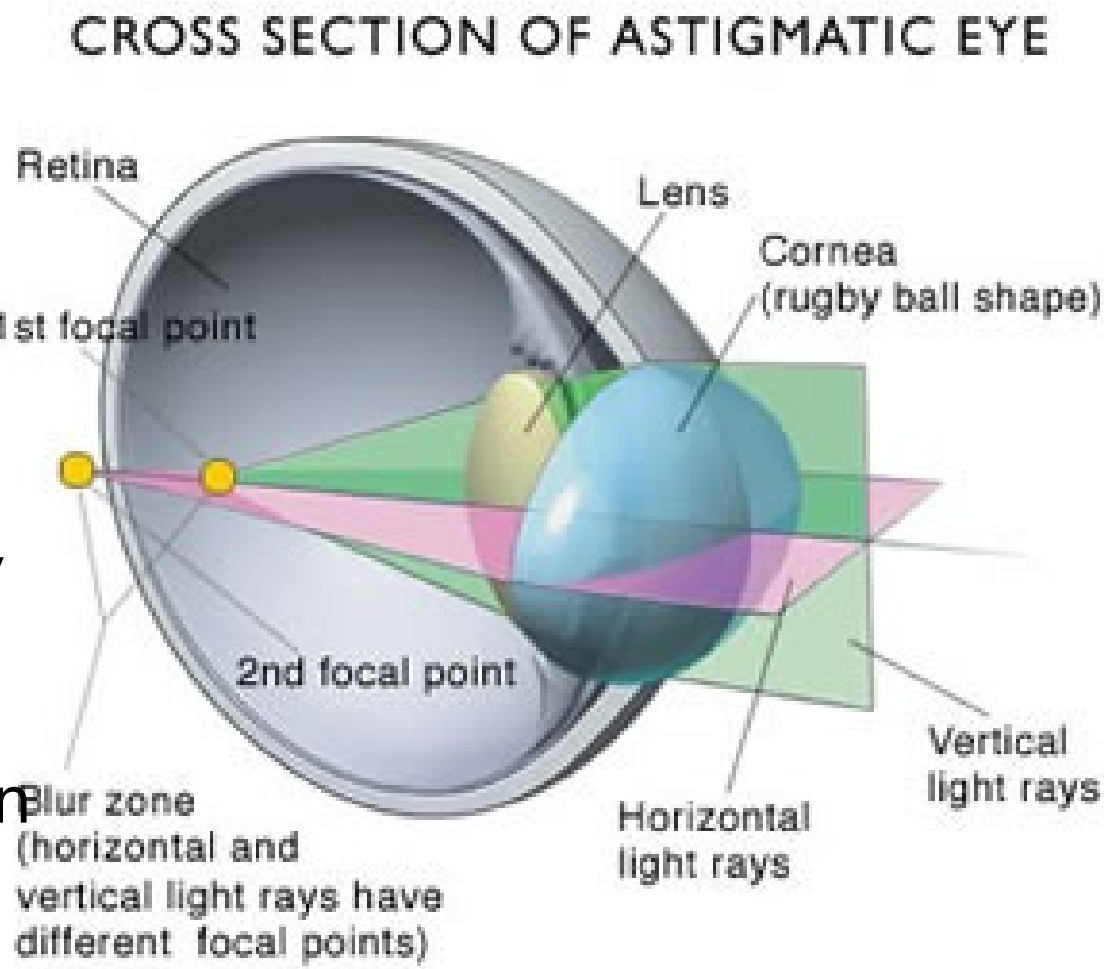
Definition:

- Lack of a focal point
- Curvature anomaly of the refractive media such that parallel incidence rays do not converge at a point but are drawn apart to form a line

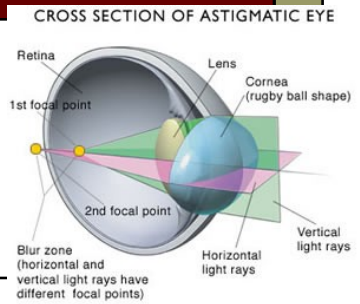


Astigmatism.....

- Definition.....:
  - Refractive media: (Ant. and Post. surface Cornea & Lens) are not spherical;
  - Refract differently along one meridian and along the meridian 90 degree to it: 2 focal points



Astigmatism……

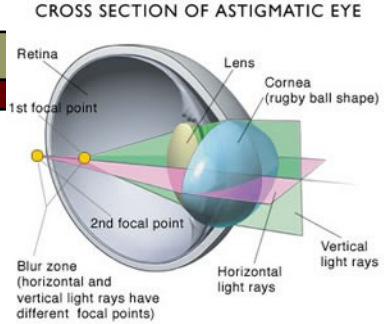


Epidemiology

- 42% of all humans $\geq \pm 0.5D$: 20% $\geq$ 1.00D

Classification

- External: (ant surface of the cornea) Internal: other components
- Rule
  - With the rule astigmatism: (most common)
    - meridian with the greatest refractive power is **Vertical (70 - 110 degrees)**
  - Against the rule astigmatism
    - meridian with the greatest refractive power is **Horizontal (160-20 degrees)**
  - Oblique astigmatism
- Regular/Irregular

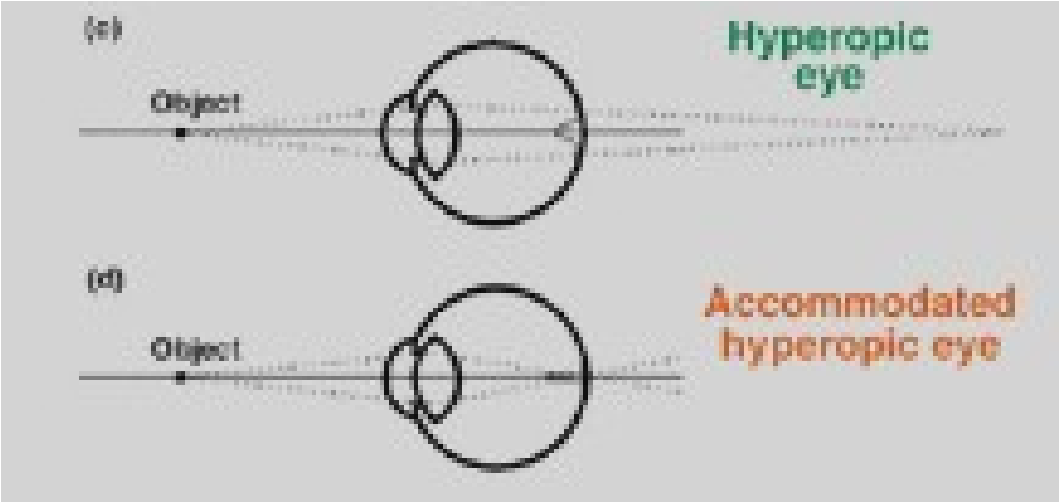


Astigmatism……

- Symptoms
  - Distorted images
  - Asthenopic symptoms
- Dignosis
  - Placidos disc
  - Corneal topography
  - Refraction: (Objective, Subjective)
- Treatment
  - Cylindrical lenses (regular astigmatism)
  - Contact lenses
    - Soft
    - Rigid
    - Toric

Presbyopia

- Definition
  - Physiological loss of accommodation with advanced age
  - Treatment – corrective spectacles (bifocal, multifocal)



LASIK

- LASIK is no different than Glasses or Contact lenses
  - It changes the way the light is focused as it enters the eye
  - By changing the shape of the cornea, the light is either converged or diverged
  - Astigmatism is corrected by making the curvature of the cornea equal in all directions - making the surface of the eye round.